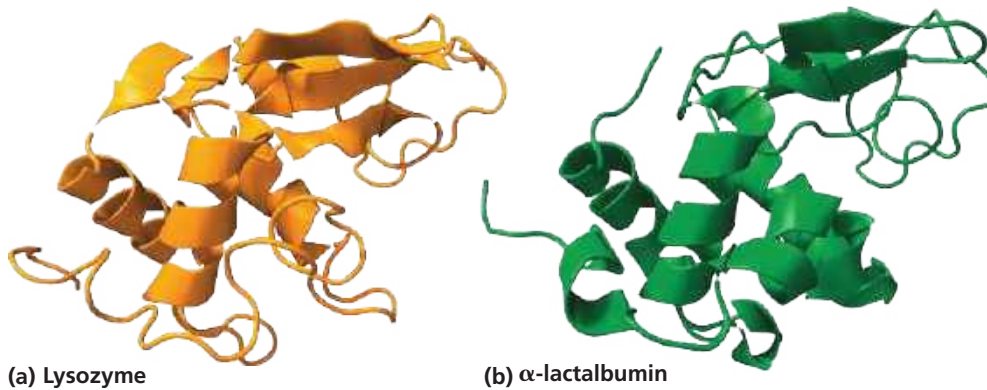


▼ **Figure 21.15 Comparison of lysozyme and α -lactalbumin proteins.** Computer-generated ribbon models of the similar structures of (a) lysozyme and (b) α -lactalbumin are shown, along with (c) a comparison of the amino acid sequences of the two proteins. Single-letter amino acid codes are used (see Figure 5.14). Identical amino acids are highlighted in yellow, and dashes indicate gaps in one sequence that have been introduced by the software to optimize the alignment.



Lysozyme	1	KVFERCELARTLKRLGMDGYRGISLANWMC	LAKWESGYNTRATN			
α -lactalbumin	1	KQFTKCELSQLLK--DIDGYGGIALPELIC	TMFHTSGYDTQAIV			
Lysozyme	51	STDYGI	FQINSRYWCNDGKTPGAVNACHLS	CSALLQDN	IADAVA	
α -lactalbumin	51	STEYGL	FQISNKLWCSSQVPQSRNICDIS	CDKFLDD	ITDDIM	
Lysozyme	101	RDPQGI	RAWVAWRNR	CQ-NRDV	RYVQGC	GV
α -lactalbumin	101	D-IKGI	DYWLAKKAL	CT--E	KLEQWL	CEKL-

(c) Amino acid sequence alignments of lysozyme and α -lactalbumin

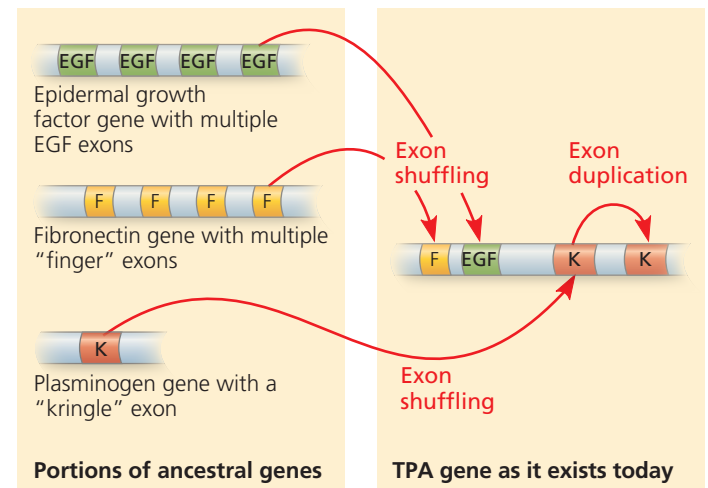
augment its function by increasing its stability, enhancing its ability to bind a particular ligand, or altering some other property. Quite a few protein-coding genes have multiple copies of related exons, which presumably arose by duplication and then diverged. The gene encoding the extracellular matrix protein collagen is a good example. Collagen is a structural protein (see Figure 5.18) with a highly repetitive amino acid sequence, which reflects the repetitive pattern of exons in the collagen gene.

As an alternative possibility, we can imagine the occasional mixing and matching of different exons either within a gene or between two different (nonallelic) genes owing to errors in meiotic recombination. This process, termed *exon shuffling*, could lead to new proteins with novel combinations of functions. As an example, let's consider the gene for tissue plasminogen activator (TPA). The TPA protein is an extracellular protein that helps control blood clotting. It has four domains of three types, each encoded by an exon, and one of those exons is present in two copies. Because each type of exon is also found in other proteins, the current version of the gene for TPA is thought to have arisen by several instances of exon shuffling during errors in meiotic recombination and subsequent duplication (Figure 21.16).

MAKE CONNECTIONS Even though two amino acids are not identical, they may be structurally and chemically similar and therefore behave similarly. Using Figure 5.14 as a reference, examine the nonidentical amino acids in positions 1–30 and note cases where the amino acids in the two sequences are similarly acidic or basic.

▼ **Figure 21.16 Evolution of a new gene by exon shuffling.**

Meiotic errors could have moved exons, each encoding a particular domain, from ancestral forms of the genes for epidermal growth factor, fibronectin, and plasminogen (left) into the evolving gene for tissue plasminogen activator, TPA (right). Subsequent duplication of the “kringle” exon (K) from the plasminogen gene after its movement into the TPA gene could account for the two copies of this exon in the TPA gene existing today.



VISUAL SKILLS Looking at Figure 21.13, describe the steps by which transposable elements within introns might have facilitated the exon shuffling shown here.